

Machine Learning

CMPSCI 589

Bruno C. da Silva

bsilva@cs.umass.edu

Naive Bayes (1/2)

UMassAmherst
College of Information
& Computer Sciences



This work is licensed under a [Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License](https://creativecommons.org/licenses/by-nc-nd/4.0/)



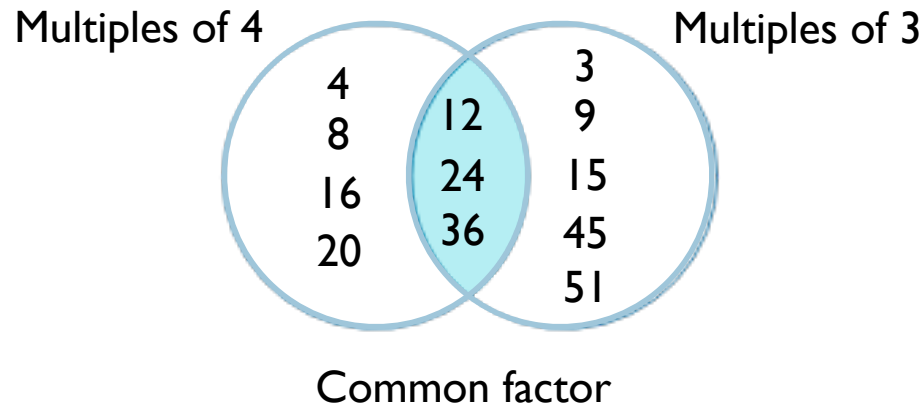
Naive Bayes Algorithm

but before that...

Probability Theory - Review



Review: Probability Theory

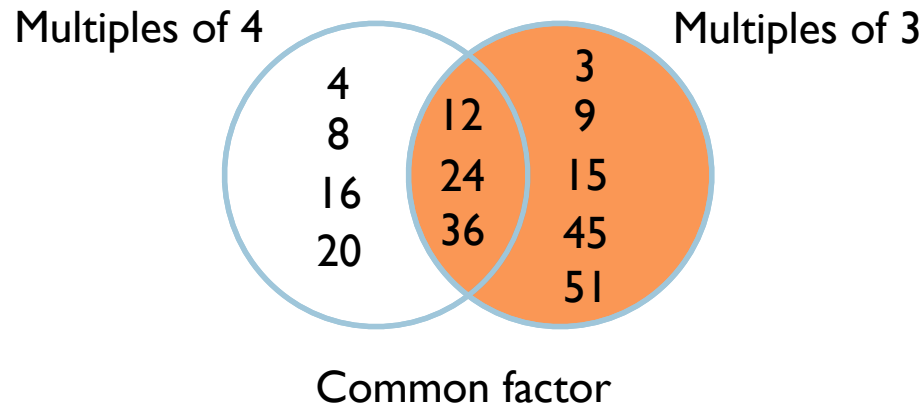


$$\Pr(\text{Multiple of 4}) = 7/12$$

$$\Pr(\text{Multiple of 3}) = 8/12$$

$$\Pr(\text{Multiple of 4 and Multiple of 3}) = 3/12$$

Review: Probability Theory



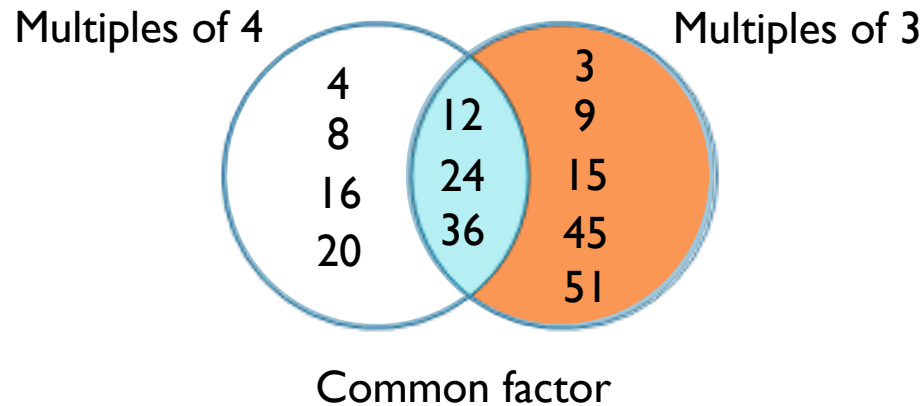
$$\Pr(\text{Multiple of 4}) = 7/12$$

$$\Pr(\text{Multiple of 3}) = 8/12$$

$$\Pr(\text{Multiple of 4 and Multiple of 3}) = 3/12$$

$$\Pr(\text{Multiple of 4} \mid \text{Multiple of 3})$$

Review: Probability Theory



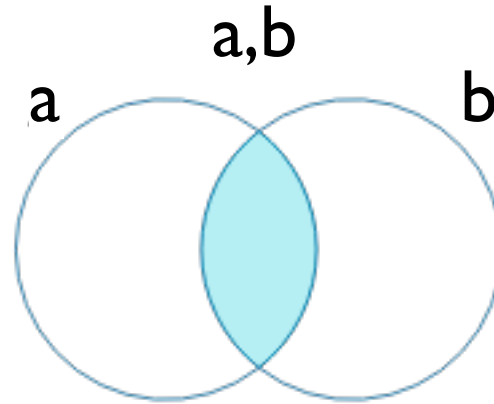
$$\Pr(\text{Multiple of 4}) = 7/12$$

$$\Pr(\text{Multiple of 3}) = 8/12$$

$$\Pr(\text{Multiple of 4 and Multiple of 3}) = 3/12$$

$$\Pr(\text{Multiple of 4} \mid \text{Multiple of 3}) = \frac{\Pr(\text{Multiple of 4 and Multiple of 3})}{\Pr(\text{Multiple of 3})}$$

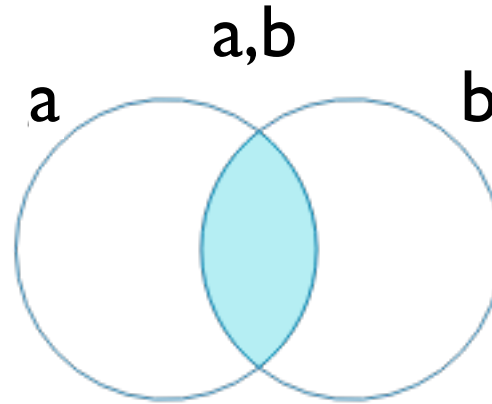
Review: Probability Theory



$$\Pr(\text{Multiple of 4} \mid \text{Multiple of 3}) = \frac{\Pr(\text{Multiple of 4 and Multiple of 3})}{\Pr(\text{Multiple of 3})}$$

$$\Pr(a \mid b) = \frac{\Pr(a, b)}{\Pr(b)}$$

Review: Probability Theory



$$\Pr(a \mid b) = \frac{\Pr(a, b)}{\Pr(b)}$$

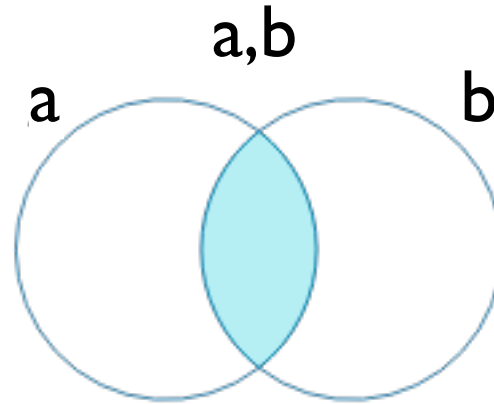
$$\Pr(b \mid a) = \frac{\Pr(b, a)}{\Pr(a)}$$

$$\Pr(a \mid b) \Pr(b) = \Pr(a, b)$$

$$\Pr(b \mid a) \Pr(a) = \Pr(b, a)$$

$$\Pr(a, b) = \Pr(b, a)$$

Review: Probability Theory



$$\Pr(a \mid b) = \frac{\Pr(a, b)}{\Pr(b)}$$

$$\Pr(b \mid a) = \frac{\Pr(b, a)}{\Pr(a)}$$

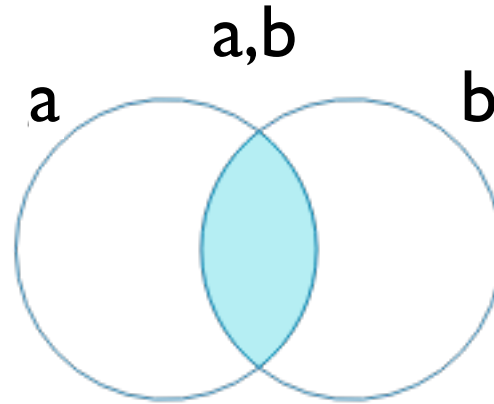
$$\Pr(a \mid b) \Pr(b) = \Pr(a, b)$$

$$\Pr(b \mid a) \Pr(a) = \Pr(a, b)$$

$$\Pr(a, b) = \Pr(b, a)$$



Review: Probability Theory



$$\Pr(a \mid b) = \frac{\Pr(a, b)}{\Pr(b)}$$

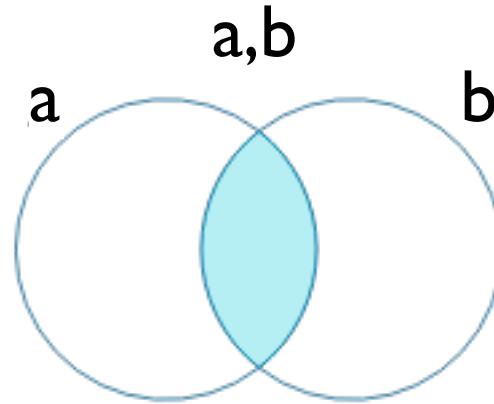
$$\Pr(b \mid a) = \frac{\Pr(b, a)}{\Pr(a)}$$

$$\Pr(a \mid b) \Pr(b) = \Pr(a, b)$$

$$\Pr(b \mid a) \Pr(a) = \Pr(a, b)$$

$$\Pr(a \mid b) \Pr(b) = \Pr(b \mid a) \Pr(a)$$

Review: Probability Theory



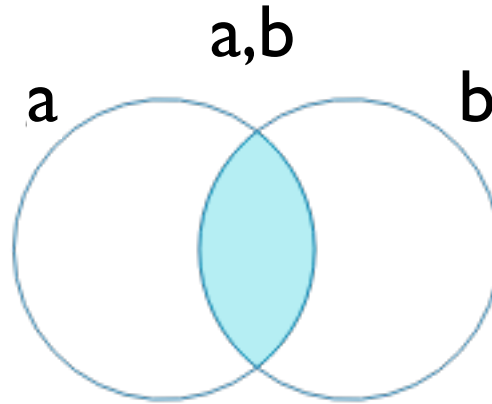
$$\Pr(a \mid b) = \frac{\Pr(a, b)}{\Pr(b)}$$

$$\Pr(b \mid a) = \frac{\Pr(b, a)}{\Pr(a)}$$

$$\Pr(a \mid b) \Pr(b) = \Pr(b \mid a) \Pr(a)$$

$$\Pr(a \mid b) = \frac{\Pr(b \mid a) \Pr(a)}{\Pr(b)}$$

Review: Probability Theory



$$\Pr(a \mid b) = \frac{\Pr(a, b)}{\Pr(b)}$$

$$\Pr(b \mid a) = \frac{\Pr(b, a)}{\Pr(a)}$$

$$\Pr(a \mid b) = \frac{\Pr(b \mid a) \Pr(a)}{\Pr(b)}$$

Bayes Theorem

Review: Bayes Theorem

$$\Pr(a | b) = \frac{\Pr(b | a) \Pr(a)}{\Pr(b)}$$

- Only 1% of the population has cancer (**C**)
- 0.2% of the population is 65 years old (**A**)
- Considering the group of people who have cancer, 0.5% are 65 years old
- If we don't know anything about John, what is the probability that he has cancer? = 1%
- If we don't know anything about John, what is the probability that he does not have cancer? = 99%
- If we know that John is 65 years old, what is the probability that he has cancer? = 2.5%
- If we know that John is 65 years old, what is the probability that he does not have cancer? = 97.5%

$$\Pr(C|A) = \frac{\Pr(A | C) \Pr(C)}{\Pr(A)} = \frac{(0.5/100) \times (1/100)}{(0.2/100)} = \frac{(0.5 \times 1)/100}{0.2} = 2.5 \%$$

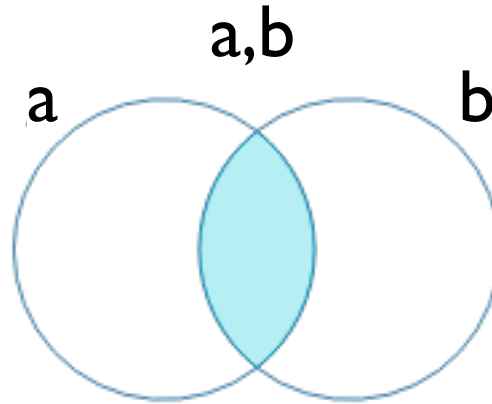
Review: Bayes Theorem

$$\Pr(a | b) = \frac{\Pr(b | a) \Pr(a)}{\Pr(b)}$$

- Only 1% of the population has cancer (**C**)
- 0.2% of the population is 65 years old (**A**)
- Considering the group of people who have cancer, 0.5% are 65 years old
- If we don't know anything about John, what is the probability that he has cancer? = 1%
- If we don't know anything about John, what is the probability that he does not have cancer? = 99%
- If we know that John is 65 years old, what is the probability that he has cancer? = 2.5%
- If we know that John is 65 years old, what is the probability that he does not have cancer? = 97.5%

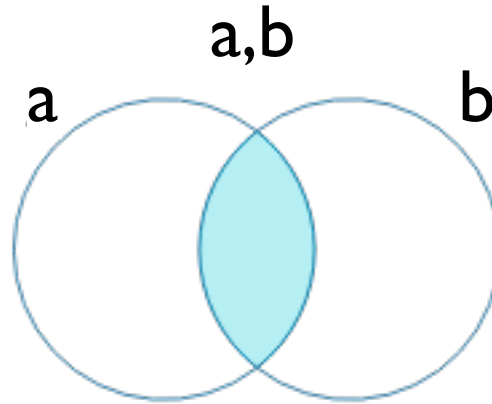
After we found out John's age (evidence),
we adjusted our prior belief (a priori probability of cancer) → prob. cancer = 1%
and obtained an updated belief (a posteriori probability) → prob. cancer = 2.5%

Review: Chain Rule



$$\Pr(a, b, c) = \Pr(a \mid b, c) \Pr(b \mid c) \Pr(c)$$

Review: Chain Rule



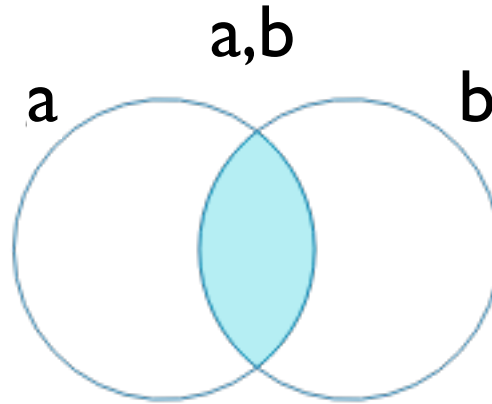
$$\Pr(a, b, c) = \Pr(a \mid b, c) \Pr(b \mid c) \Pr(c)$$

Similarly...

$$\Pr(a, b, c, d) = \Pr(a \mid b, c, d) \Pr(b \mid c, d) \Pr(c \mid d) \Pr(d)$$

Chain Rule

Review: Conditional Independence



$$\Pr(a \mid \bullet, c) = \Pr(a \mid c)$$

*If a and any $\bullet$ are independent given c
we can "delete" $\bullet$ from the conditional part of the equation*

$$\Pr(\text{go to Bluewall} \mid \text{sunny in Japan, hungry}) = \Pr(\text{go to Bluewall} \mid \text{hungry})$$

*Can "delete" **sunny in Japan** because I am conditioning
on what matters (**hungry**)*

Review: Conditional Independence

$$\Pr(a \mid b) = \frac{\Pr(b \mid a) \Pr(a)}{\Pr(b)} \quad \textbf{Bayes Theorem}$$

$$\Pr(a, b, c, d) = \Pr(a \mid b, c, d) \Pr(b \mid c, d) \Pr(c \mid d) \Pr(d)$$

Chain Rule

$$\Pr(a \mid \bullet, c) = \Pr(a \mid c)$$

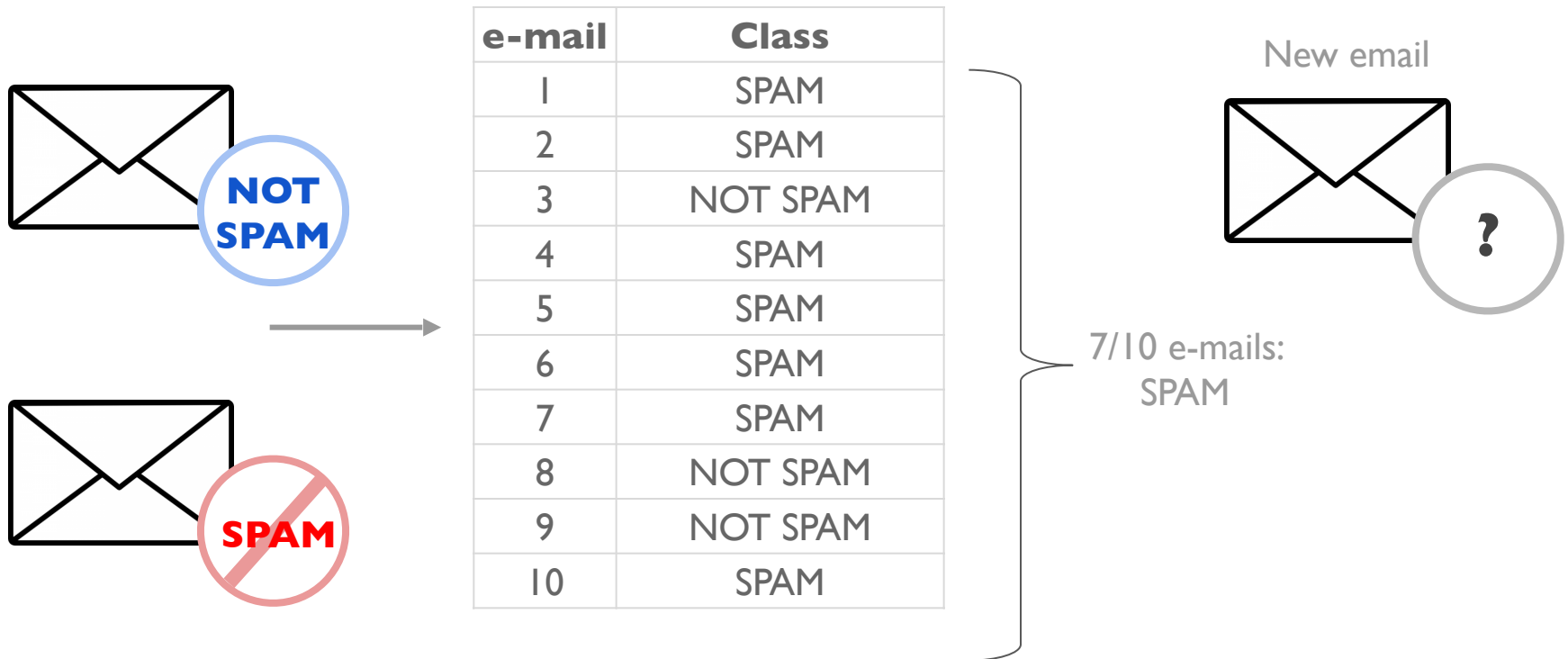
If a and any $\bullet$ are independent given c

we can "delete" $\bullet$ from the conditional part of the equation

Naive Bayes Algorithm

Most Likely Class/Label

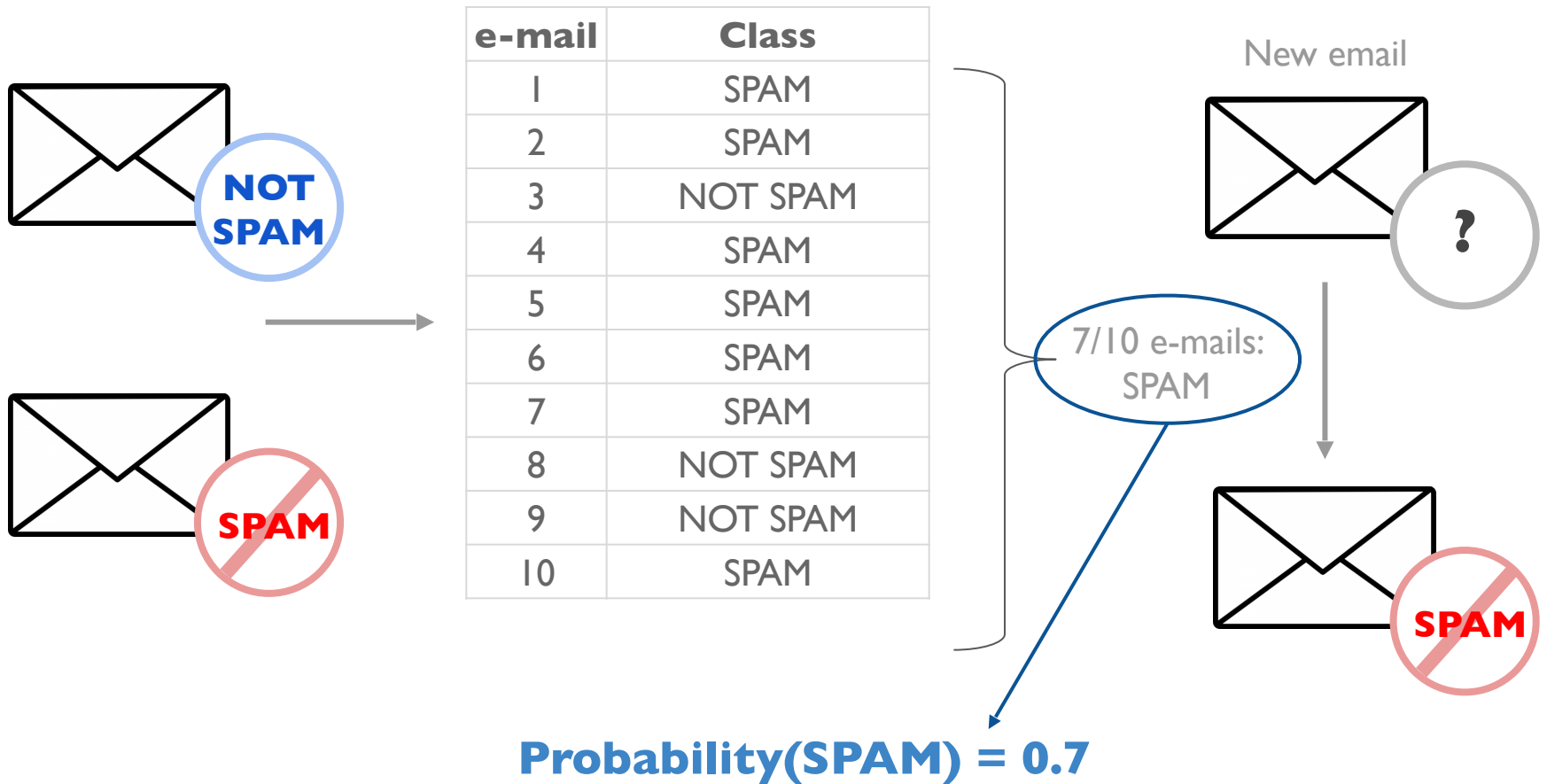
- Intuition



“Given a new email, is it more likely to be a SPAM or NOT SPAM?”

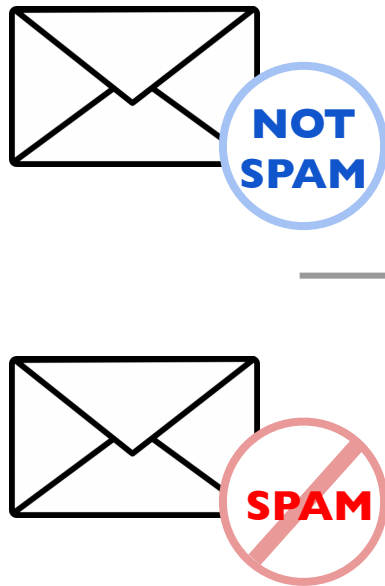
Most Likely Class/Label

- Intuition

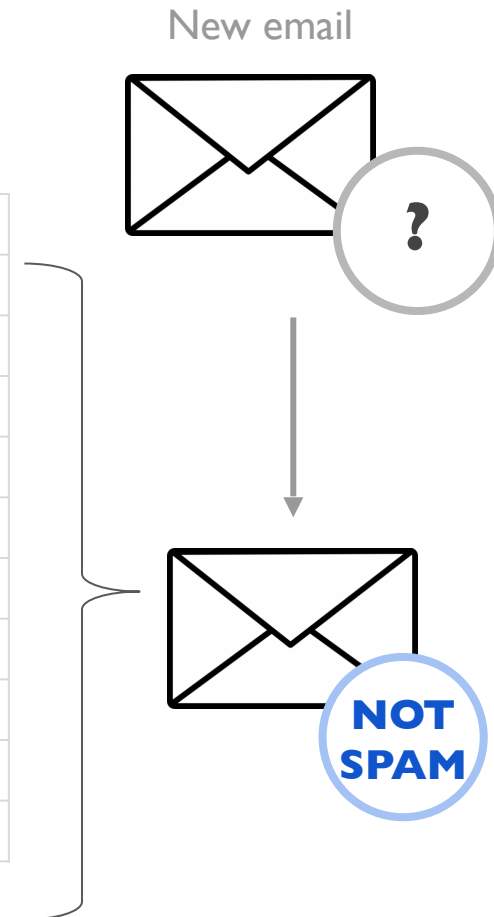


Most Likely Class/Label

- Intuition



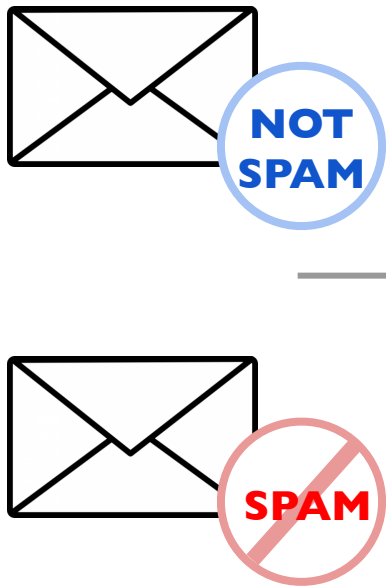
e-mail	Contents	Class
1	"password expired"	SPAM
2	"send password"	SPAM
3	"review conference"	NOT SPAM
4	"password review"	SPAM
5	"review account"	SPAM
6	"account password"	SPAM
7	"send account"	SPAM
8	"conference paper"	NOT SPAM
9	"send paper"	NOT SPAM
10	"expired account"	SPAM



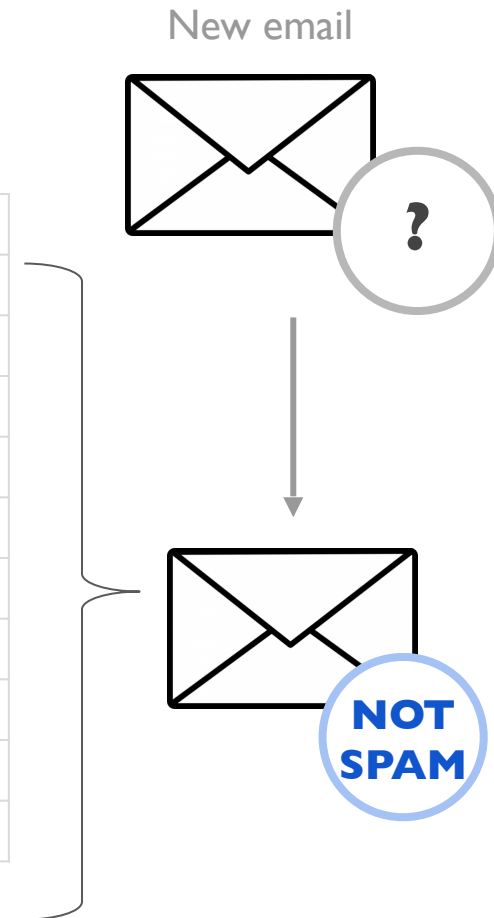
“Given a new email whose contents are *"review conference paper"*, it is more likely to be a SPAM or NOT SPAM?

Most Likely Class/Label

- Intuition



e-mail	Contents	Class
1	"password expired"	SPAM
2	"send password"	SPAM
3	"review conference"	NOT SPAM
4	"password review"	SPAM
5	"review account"	SPAM
6	"account password"	SPAM
7	"send account"	SPAM
8	"conference paper"	NOT SPAM
9	"send paper"	NOT SPAM
10	"expired account"	SPAM



What is the most likely class of the email?

Naive Bayes Classifier

- Probabilistic classifier
- Computes the probability of each class/label based on input attributes of an instance

$$\Pr(y_1 | \mathbf{x})$$

$$\Pr(y_2 | \mathbf{x})$$

...

$$\Pr(y_N | \mathbf{x})$$

} max

$$\Pr(y_i | \mathbf{x}) = \Pr(y_i | x_1, \dots, x_n)$$

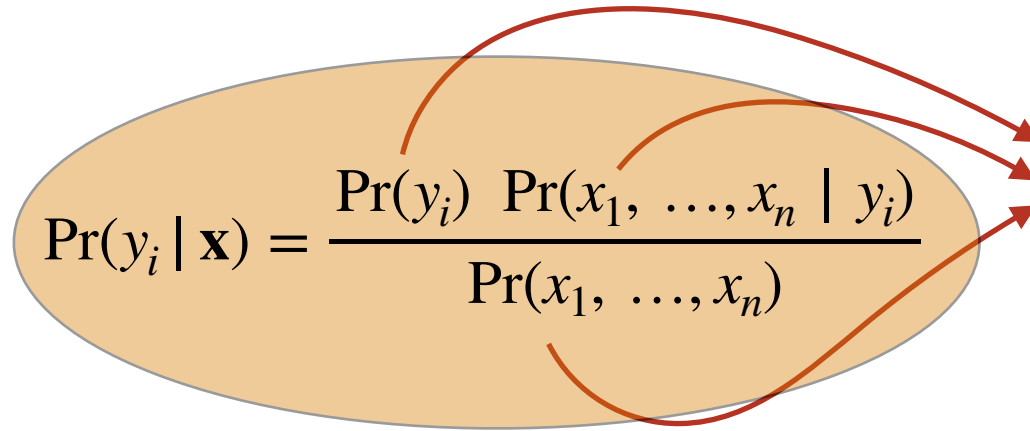
$$C = \arg \max_{y_i \in Y} \Pr(y_i | x_1, \dots, x_n)$$

Bayes Theorem

Naive Bayes Classifier

$$\Pr(y_i | \mathbf{x}) = \Pr(y_i | x_1, \dots, x_n) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

Naive Bayes Classifier


$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

The equation is enclosed in a light orange oval. Four red curved arrows originate from the terms $\Pr(y_i)$, $\Pr(x_1, \dots, x_n | y_i)$, $\Pr(x_1, \dots, x_n)$, and the entire fraction, pointing towards the text box on the right.

How to compute
these probabilities
based on
a training set?

y_1 = spam

y_2 = not_spam

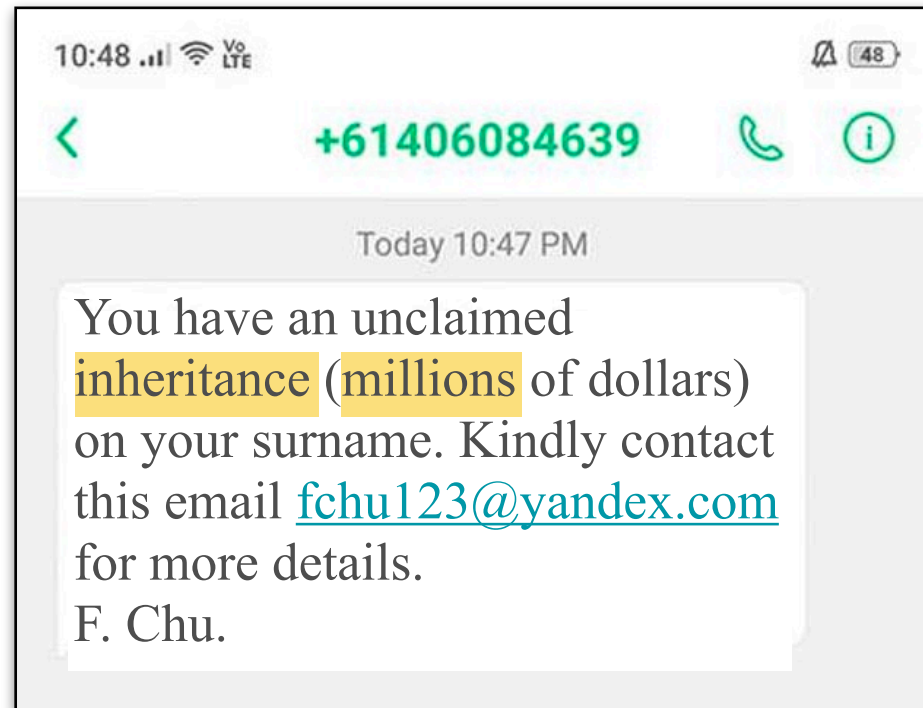
$\mathbf{x} = (x_1, \dots, x_n)$ = words in email

Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$



Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) = \frac{\Pr(y = \text{spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{spam})}{\Pr(x_1 = \text{inheritance}, x_2 = \text{millions})}$$

$$\Pr(y = \text{not_spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) = \frac{\Pr(y = \text{not_spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{not_spam})}{\Pr(x_1 = \text{inheritance}, x_2 = \text{millions})}$$

Which class has the highest probability, given the contents of the email?

Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) = \frac{\Pr(y = \text{spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{spam})}{\Pr(x_1 = \text{inheritance}, x_2 = \text{millions})}$$

$$\Pr(y = \text{not_spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) = \frac{\Pr(y = \text{not_spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{not_spam})}{\Pr(x_1 = \text{inheritance}, x_2 = \text{millions})}$$

IF $\Pr(\text{spam} | \text{contents}) > \Pr(\text{not_spam} | \text{contents})$



Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) = \frac{\Pr(y = \text{spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{spam})}{\Pr(x_1 = \text{inheritance}, x_2 = \text{millions})}$$

$$\Pr(y = \text{not_spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) = \frac{\Pr(y = \text{not_spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{not_spam})}{\Pr(x_1 = \text{inheritance}, x_2 = \text{millions})}$$

IF $\Pr(\text{spam} | \text{contents}) < \Pr(\text{not_spam} | \text{contents})$



Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

same denominator in both
quantities being compared
→ can be ignored

$$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) = \frac{\Pr(y = \text{spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{spam})}{\Pr(x_1 = \text{inheritance}, x_2 = \text{millions})}$$

$$\Pr(y = \text{not_spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) = \frac{\Pr(y = \text{not_spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{not_spam})}{\Pr(x_1 = \text{inheritance}, x_2 = \text{millions})}$$

Which class has the highest probability, given the contents of the email?

Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto \Pr(y = \text{spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{spam})$$

$$\Pr(y = \text{not_spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto \Pr(y = \text{not_spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{not_spam})$$

Which class has the highest probability, given the contents of the email?

Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto \Pr(y = \text{spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{spam})$$

$$\Pr(y = \text{not_spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto \Pr(y = \text{not_spam}) \Pr(x_1 = \text{inheritance}, x_2 = \text{millions} | y = \text{not_spam})$$

Which class has the highest probability, given the contents of the email?

$$\arg \max_{y_i \in Y} \Pr(y_i | x_1, x_2, \dots, x_n)$$

Naive Bayes Classifier

$$\arg \max_{y_i \in Y} \Pr(y_i \mid x_1, x_2, \dots, x_n)$$

$$\arg \max_{y_i \in Y} \frac{\Pr(x_1, x_2, \dots, x_n \mid y_i) \Pr(y_i)}{\Pr(x_1, x_2, \dots, x_n)}$$

$$\Pr(x_1, x_2, \dots, x_n \mid y_i) \text{ ?}$$

Remember the chain rule:

$$\Pr(a, b, c, d) = \Pr(a \mid b, c, d) \Pr(b \mid c, d) \Pr(c \mid d) \Pr(d)$$

$$\Pr(x_1, x_2, \dots, x_n \mid y_i) =$$

$$\Pr(x_1 \mid x_2, x_3, \dots, x_n, y_i) \Pr(x_2 \mid x_3, x_4, \dots, x_n, y_i) \dots \Pr(x_{n-1} \mid x_n, y_i) \Pr(x_n \mid y_i)$$

Naive Bayes Classifier

$$\arg \max_{y_i \in Y} \frac{\Pr(x_1, x_2, \dots, x_n \mid y_i) \Pr(y_i)}{\Pr(x_1, x_2, \dots, x_n)}$$

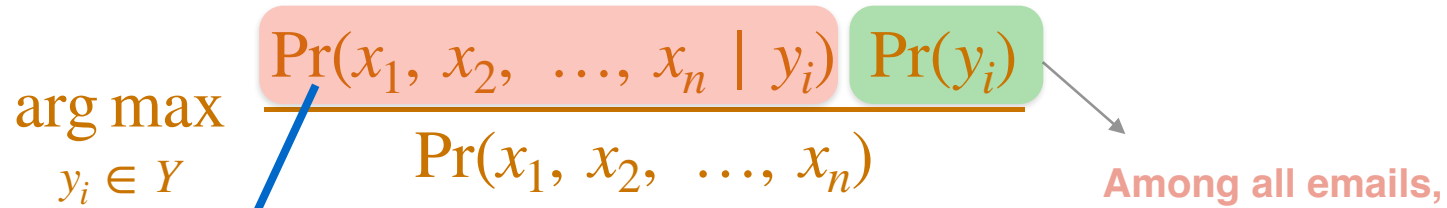
Easy/fast to count

Among all emails, the percentage of spams

$$\Pr(x_1, x_2, \dots, x_n \mid y_i) =$$

$$\Pr(x_1 \mid x_2, x_3, \dots, x_n, y_i) \Pr(x_2 \mid x_3, x_4, \dots, x_n, y_i) \dots \Pr(x_{n-1} \mid x_n, y_i) \Pr(x_n \mid y_i)$$

Naive Bayes Classifier

$$\arg \max_{y_i \in Y} \frac{\Pr(x_1, x_2, \dots, x_n \mid y_i) \Pr(y_i)}{\Pr(x_1, x_2, \dots, x_n)}$$


Among all emails,
the percentage
of spams

Easy/fast to count

$$\Pr(x_1, x_2, \dots, x_n \mid y_i) =$$

$$\Pr(x_1 \mid x_2, x_3, \dots, x_n, y_i) \Pr(x_2 \mid x_3, x_4, \dots, x_n, y_i) \dots \Pr(x_{n-1} \mid x_n, y_i) \Pr(x_n \mid y_i)$$

(example)

Naive Bayes Classifier

$$\arg \max_{y_i \in Y} \frac{\Pr(x_1, x_2, \dots, x_n \mid y_i) \Pr(y_i)}{\Pr(x_1, x_2, \dots, x_n)}$$

Among all emails, the percentage of spams

Easy/fast to count

$$\Pr(x_1, x_2, \dots, x_n \mid y_i) =$$

$$\Pr(x_1 \mid x_2, x_3, \dots, x_n, y_i) \Pr(x_2 \mid x_3, x_4, \dots, x_n, y_i) \dots \Pr(x_{n-1} \mid x_n, y_i) \Pr(x_n \mid y_i)$$

$$\Pr(x_1 = \text{inheritance}, x_2 = \text{millions} \mid y_i = \text{spam}) = \text{[using the Chain Rule...]}$$

$$\Pr(x_1 = \text{inheritance} \mid x_2 = \text{millions}, y_i = \text{spam}) \Pr(x_2 = \text{millions} \mid y_i = \text{spam})$$

Naive Bayes Classifier

$$\arg \max_{y_i \in Y} \frac{\Pr(x_1, x_2, \dots, x_n \mid y_i) \Pr(y_i)}{\Pr(x_1, x_2, \dots, x_n)}$$

Among all emails, the percentage of spams

Easy/fast to count

$$\Pr(x_1, x_2, \dots, x_n \mid y_i) =$$

$$\Pr(x_1 \mid x_2, x_3, \dots, x_n, y_i) \Pr(x_2 \mid x_3, x_4, \dots, x_n, y_i) \dots \Pr(x_{n-1} \mid x_n, y_i) \Pr(x_n \mid y_i)$$

$$\Pr(x_1 = \text{inheritance}, x_2 = \text{millions} \mid y_i = \text{spam}) = [\text{using the Chain Rule...}]$$

$$\Pr(x_1 = \text{inheritance} \mid x_2 = \text{millions}, y_i = \text{spam}) \Pr(x_2 = \text{millions} \mid y_i = \text{spam})$$

Among the spams, the percentage that contain the word *millions*

Easy/fast to count

Naive Bayes Classifier

$$\arg \max_{y_i \in Y} \frac{\Pr(x_1, x_2, \dots, x_n \mid y_i) \Pr(y_i)}{\Pr(x_1, x_2, \dots, x_n)}$$

Among all emails, the percentage of spams

$$\Pr(x_1, x_2, \dots, x_n \mid y_i) =$$

Easy/fast to count

$$\Pr(x_1 \mid x_2, x_3, \dots, x_n, y_i) \Pr(x_2 \mid x_3, x_4, \dots, x_n, y_i) \dots \Pr(x_{n-1} \mid x_n, y_i) \Pr(x_n \mid y_i)$$

$$\Pr(x_1 = \text{inheritance}, x_2 = \text{millions} \mid y_i = \text{spam}) = [\text{using the Chain Rule...}]$$

$$\Pr(x_1 = \text{inheritance} \mid x_2 = \text{millions}, y_i = \text{spam}) \Pr(x_2 = \text{millions} \mid y_i = \text{spam})$$

Among the spams that contain the word *millions*, the percentage that also contain the word *inheritance*

Hard/slow to count

Among the spams, the percentage that contain the word *millions*

Easy/fast to count

Naive Bayes Classifier

$$\Pr(x_1 = \text{inheritance}, x_2 = \text{millions} \mid y_i = \text{spam}) = [\text{using the Chain Rule...}]$$

$$\Pr(x_1 = \text{inheritance} \mid x_2 = \text{millions}, y_i = \text{spam}) \Pr(x_2 = \text{millions} \mid y_i = \text{spam})$$

Among the **spams** that contain the word **millions**,
the percentage that also contain the word **inheritance**

Hard/slow to count

Naive Bayes assumption:
inheritance and **millions** are conditionally independent given **spam**

That is: if I know the email is a **spam**,
knowing that it contains the word **inheritance**
does not tell me anything
about how likely it is for **millions** to also show up

Naive Bayes Classifier

$$\Pr(x_1 = \text{inheritance}, x_2 = \text{millions} \mid y_i = \text{spam}) = [\text{using the Chain Rule...}]$$

$$\Pr(x_1 = \text{inheritance} \mid x_2 = \text{millions}, y_i = \text{spam}) \Pr(x_2 = \text{millions} \mid y_i = \text{spam})$$

Among the **spams** that contain the word **millions**,
the percentage that also contain the word **inheritance**

Hard/slow to count

Naive Bayes assumption:
inheritance* and *millions* are conditionally independent given *spam

The probability of ***inheritance*** and ***millions*** are independent given that it's a ***spam***

The probability of ***inheritance*** showing up depends **only** on whether it is a spam
but it does **not** depend on any other words

Naive Bayes Classifier

$$\Pr(x_1 = \text{inheritance}, x_2 = \text{millions} \mid y_i = \text{spam}) = [\text{using the Chain Rule...}]$$

$$\Pr(x_1 = \text{inheritance} \mid x_2 = \text{millions}, y_i = \text{spam}) \Pr(x_2 = \text{millions} \mid y_i = \text{spam})$$

Among the **spams** that contain the word **millions**,
the percentage that also contain the word **inheritance**

Hard/slow to count

Naive Bayes assumption:
inheritance and **millions** are conditionally independent given **spam**

True? Not really.... these words show up together all the time!

$$\Pr(x_1 = \text{inheritance} \mid x_2 = \text{millions}, y_i = \text{spam}) =$$

Naive Bayes Classifier

Naive Bayes assumption:
inheritance and *millions* are conditionally independent given *spam*

True? Not really.... these words show up together all the time!

$$\Pr(x_1 = \text{inheritance} \mid x_2 = \text{millions}, y_i = \text{spam}) = \Pr(x_1 = \text{inheritance} \mid y_i = \text{spam})$$

Naive Bayes Classifier

Let's skip the theory / full derivation of the algorithm (for now)

So you see what we'll end up getting → Very simple algorithm!

Naive Bayes assumption:
inheritance and *millions* are conditionally independent given *spam*

True? Not really.... these words show up together all the time!

$$\Pr(x_1 = \text{inheritance} \mid x_2 = \text{millions}, y_i = \text{spam}) = \Pr(x_1 = \text{inheritance} \mid y_i = \text{spam})$$

Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto \Pr(y = \text{spam}) \times \\ \Pr(x_1 = \text{inheritance} | y = \text{spam}) \Pr(x_2 = \text{millions} | y = \text{spam})$$

$$\Pr(y = \text{not_spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto \Pr(y = \text{not_spam}) \times \\ \Pr(x_1 = \text{inheritance} | y = \text{not_spam}) \Pr(x_2 = \text{millions} | y = \text{not_spam})$$

Which class has the highest probability, given the contents of the email?

Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto \Pr(y = \text{spam}) \times$

$\Pr(x_1 = \text{inheritance} | y = \text{spam}) \Pr(x_2 = \text{millions} | y = \text{spam})$

Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto \Pr(y = \text{spam}) \times \Pr(x_1 = \text{inheritance} | y = \text{spam}) \Pr(x_2 = \text{millions} | y = \text{spam})$$

$$\frac{\text{spams}}{\text{emails}} \times \frac{\text{spams containing inheritance}}{\text{spams}} \times \frac{\text{spams containing millions}}{\text{spams}}$$

Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto \Pr(y = \text{spam}) \times \Pr(x_1 = \text{inheritance} | y = \text{spam}) \Pr(x_2 = \text{millions} | y = \text{spam})$$

$$\frac{\text{spams}}{\text{emails}} \times \frac{\text{spams containing inheritance}}{\text{spams}} \times \frac{\text{spams containing millions}}{\text{spams}}$$

Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto \Pr(y = \text{spam}) \times$

$\Pr(x_1 = \text{inheritance} | y = \text{spam}) \Pr(x_2 = \text{millions} | y = \text{spam})$

$$\frac{\text{spams}}{\text{emails}} \times \frac{\text{spams containing inheritance}}{\text{spams}} \times \frac{\text{spams containing millions}}{\text{spams}}$$

Naive Bayes Classifier

$y_1 = \text{spam}$
 $y_2 = \text{not_spam}$

$\mathbf{x} = (x_1, \dots, x_n) = \text{words in email}$

$$\Pr(y_i | \mathbf{x}) = \frac{\Pr(y_i) \Pr(x_1, \dots, x_n | y_i)}{\Pr(x_1, \dots, x_n)}$$

$\Pr(y = \text{spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto$

$$\frac{\text{spams}}{\text{emails}} \times \frac{\text{spams containing inheritance}}{\text{spams}} \times \frac{\text{spams containing millions}}{\text{spams}}$$

$\Pr(y = \text{not_spam} | x_1 = \text{inheritance}, x_2 = \text{millions}) \propto$

$$\frac{\text{not_spams}}{\text{emails}} \times \frac{\text{not_spams containing inheritance}}{\text{not_spams}} \times \frac{\text{not_spams containing millions}}{\text{not_spams}}$$

in practice



Machine Learning

CMPSCI 589

Bruno C. da Silva

bsilva@cs.umass.edu

Naive Bayes (1/2)

UMassAmherst
College of Information
& Computer Sciences



This work is licensed under a [Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License](https://creativecommons.org/licenses/by-nc-nd/4.0/)

